

Technical Document #526: Cybersecurity - Comprehensive Overview

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Multi-factor authentication (MFA) requires multiple forms of verification. It significantly reduces the risk of unauthorized access.

Zero trust architecture assumes no implicit trust and verifies every request. It minimizes the attack surface and limits lateral movement.

Encryption converts plaintext into ciphertext to protect data confidentiality. AES and RSA are widely used encryption algorithms.

Network segmentation divides networks into smaller segments. It limits the spread of attacks and improves security monitoring.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Key Takeaways:

- Incident response plans define how organizations respond to security incidents.
- Multi-factor authentication (MFA) requires multiple forms of verification.
- Network segmentation divides networks into smaller segments.